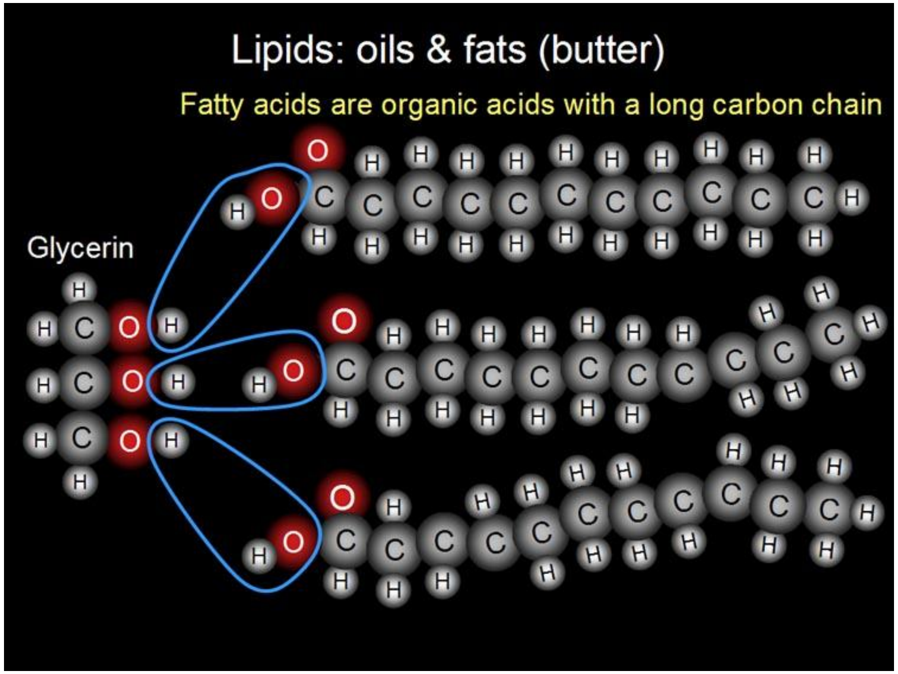
LIPIDS



DR.HUMAIRA AMAN ALI

OBJECTIVES

- At the end of the lecture student would be able to :

a. Define lipids.
b. Describe classification of lipids.
c. Understand the biological significance of lipids.

WHAT ARE LIPIDS?

- The lipids constitute a very important heterogeneous group of organic substances, including fats, oils, steroids, waxes, and related compounds.

- Related to fatty acids.

- Chemically they are various types of esters of different alcohols.

- Formed of long-chain hydrocarbon groups, also contain oxygen, phosphorus, nitrogen and sulfur.

- Utilized by living organisms.

BLOOR'S CRITERIA

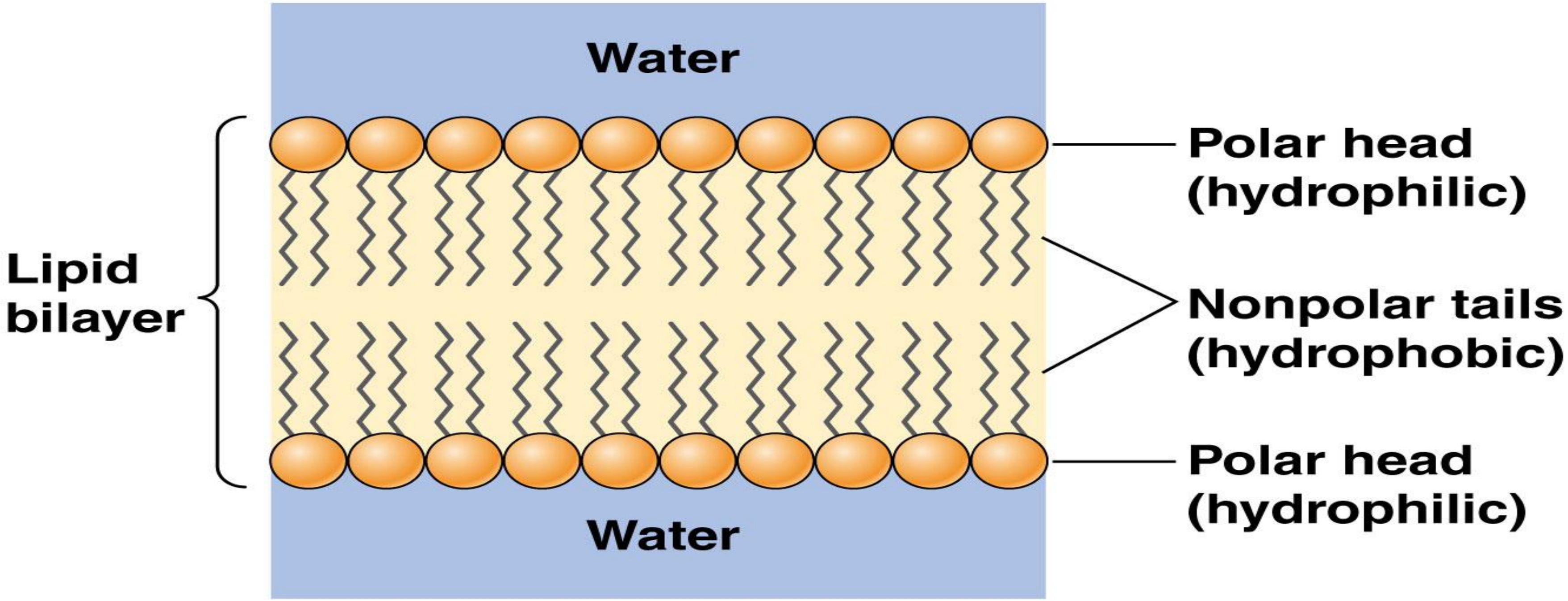
- **According to Bloor:**
  - **They are "INSOLUBLE IN WATER"**

'"SOLUBLE IN FAT SOLVENTS", Such as ether, chloroform, benzene, acetone, etc.

Lipids are non-polar (hydrophobic) compounds

BUT

Lipid molecules in cell membranes are <u>amphipathic</u> (or amphiphilic)—that is, they have a hydrophilic ("water-loving") or *polar* end and a hydrophobic ("water-fearing") or *nonpolar* end.


Water
Polar head
(hydrophilic)
Lipid
bilayer
Nonpolar tails
(hydrophobic)
Polar head
(hydrophilic)
Water

Functions of Lipids or Biomedical Importance of Lipids

The functions of lipids are given below:

- They are good sources of energy i:e 9.1 kilicalories per gram fat utilized in the body.
- Lipids in food also act as carriers of fat soluble vitamins and essential fatty acids.
- They also make food more palatable and serve to decrease its mass.
- The dietary lipids decrease gastric motility and secretions and have a high satiety value.
- Body fat provides contour to the body and also gives anatomical stability to organs like kidneys .

- When a person loses weight rapidly ,his kidneys are liable to become floating kidneys.
- Fats are good energy reservoirs in the body. Adipose tissue is best suited for this purpose.
- Lipids exert an insulating effect on the nervous tissue.
- Lipids are an integral part of cell membranes.
- Some lipids act as precursors of very important physiological compounds . For example , cholesterol is the precursor of steroid hormones while 7-dehydrocholesterol is the precursor of vitamin D3.

- Also these lipids help to prevent water evaporation from the skin ; with out this protection the amount of the daily insensible perspiration would probably be 15 to 20 liters instead of the usual 300 to 400 ml.

- The sphingosine – containing lipid have a role in the transmission of the nerve impulses across synapses.

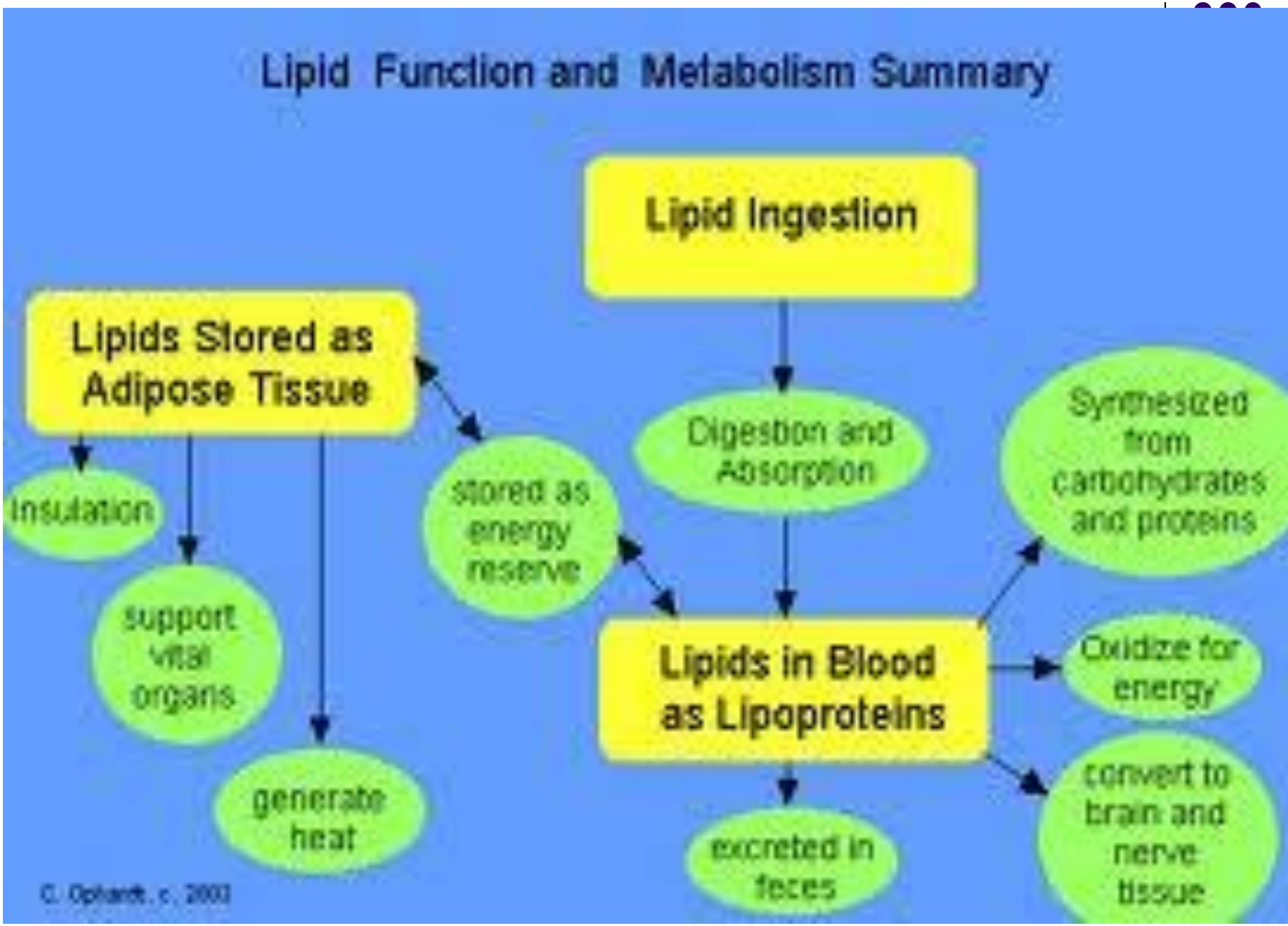
- Prostaglandins and their related compounds have hormone like actions in the body and very important physiological compounds.

- In certain cases derivatives of lipids act as intracellular messengers after they are released .

- Concerned with cell recognition, species specificity and tissue immunity.

- Bile salts are derived lipids that help the digestion and absorption of lipids.

- Lipoproteins (complexes of lipids and proteins) helps in the transport of lipids in plasma.


Lipid Function and Metabolism Summary
Lipid Ingestion
Digestion and Absorption
Lipids Stored as Adipose Tissue
insulation
support vital organs
generate heat
stored as energy reserve
Synthesized from carbohydrates and proteins
Lipids in Blood as Lipoproteins
Oxidize for energy
convert to brain and nerve tissue
excreted in feces
C. Ophardt, c. 2003

Clinical significance of lipids

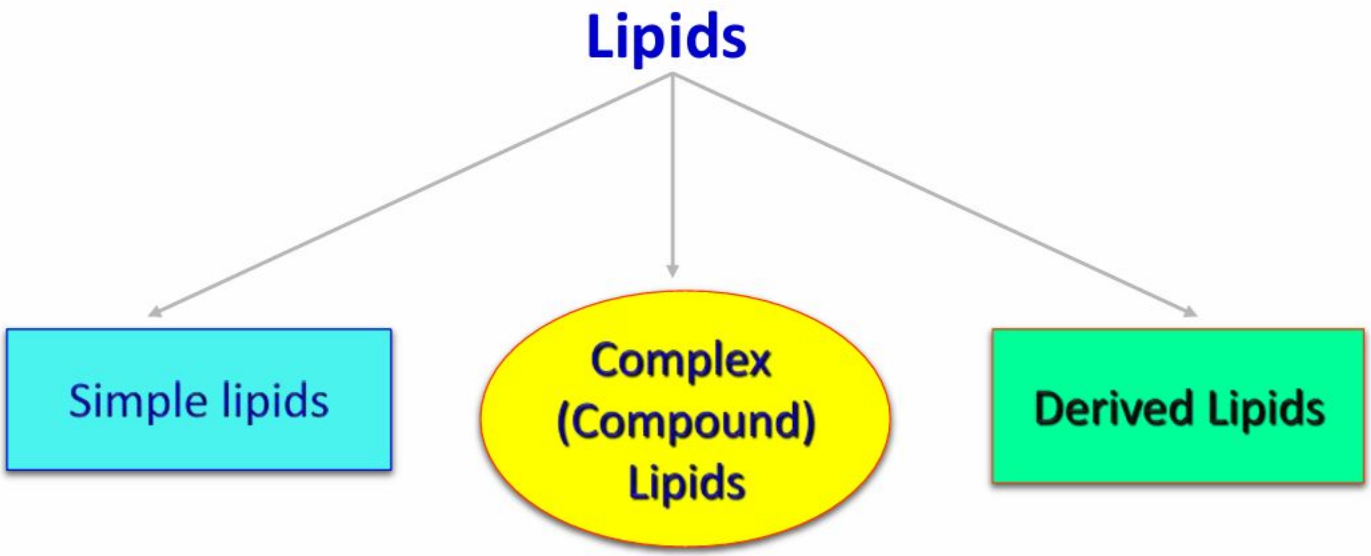
Following diseases are associated with abnormal chemistry or metabolism of lipids-

- **Obesity**
- **Atherosclerosis**
- **Diabetes Mellitus**
- **Hyperlipoproteinemia**
- **Fatty liver**
- **Lipid storage diseases**

CLASSIFICATION OF LIPIDS

- They are broadly classified based on their chemical composition into :

1. Simple lipids
2. Compound lipids
3. Derived or associated lipids

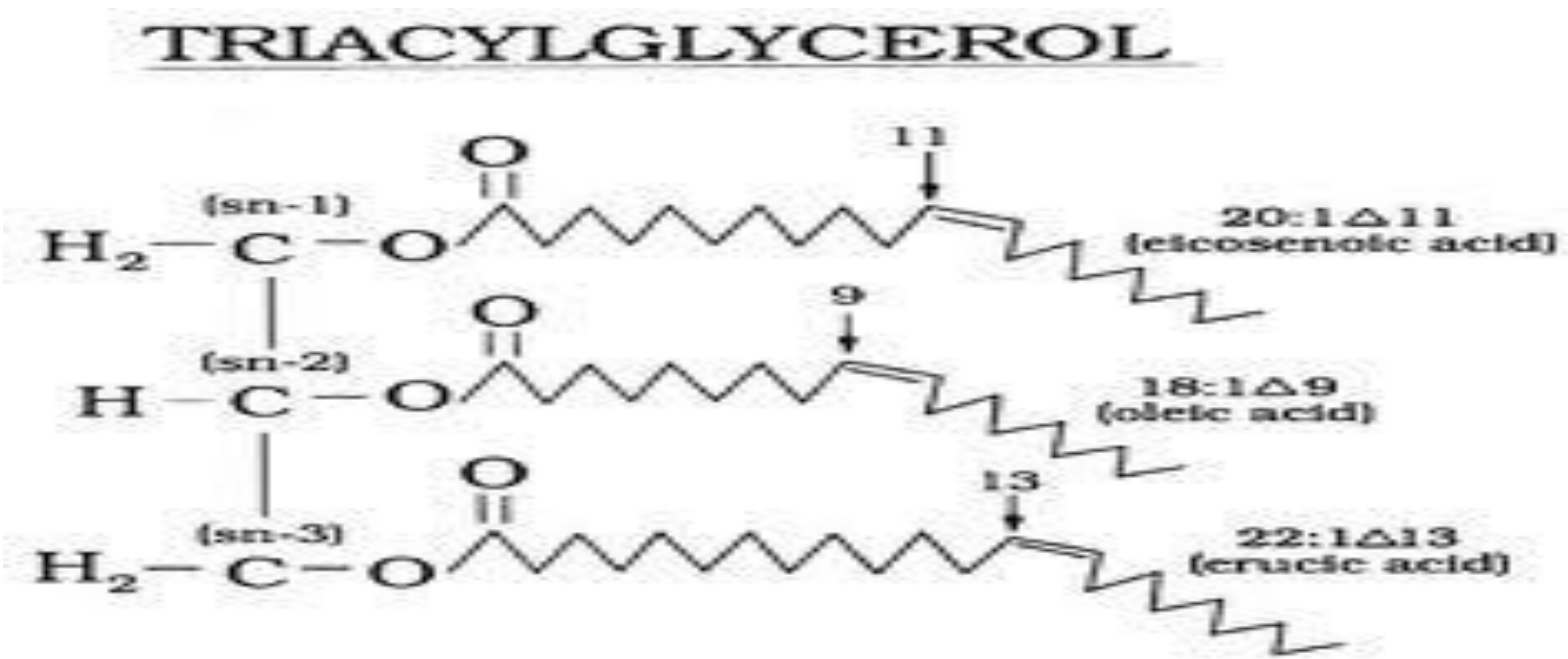


SIMPLE LIPIDS

- Simple lipids contain C, H, and O.
- These consist of following subgroups:

- **1:FATS :**
- Synonyms: Neutral fat , Triglyceride (TAGs) or Triacylglycerol.
- Def: These are esters of fatty acids with glycerol.

- **Fats** and **oils** are made from two kinds of molecules: **glycerol** (a type of alcohol with a hydroxyl group on each of its three carbons) and three **fatty acids** . Since there are three fatty acids attached, these are known as **triglycerides**.
- The main distinction between fats and oils is whether they're solid or liquid at room temperature, and this, is based on differences in the structures of the fatty acids they contain.



Properties of TAGs

- Hydrolysis:
  - TAGs undergo enzymatic hydrolysis to finally liberate free fatty acid and glycerol.
  - The process of hydrolysis, catalyzed by lipases is important for digestion of fat in GIT and fat mobilization from the adipose tissue.

- Saponification: The hydrolysis of TAG by alkali to produce glycerol and soaps is known as saponification

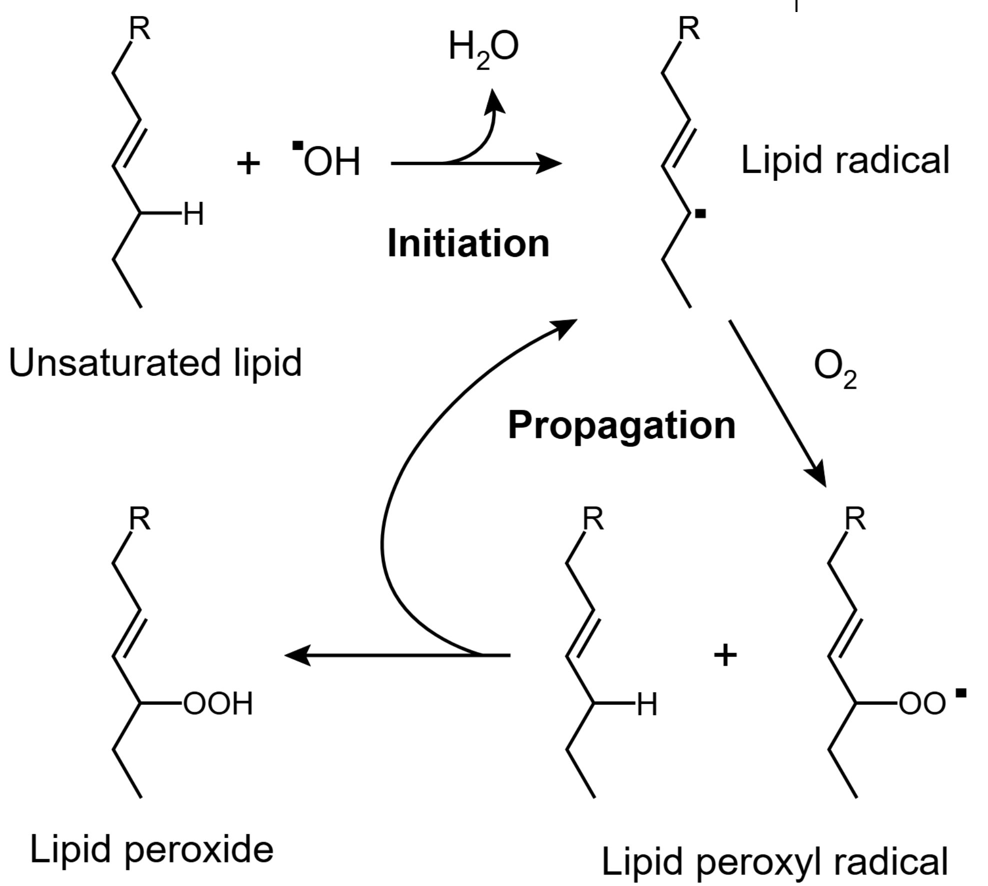
$$\text{Triacylglycerol} + 3NaOH \longrightarrow \text{Glycerol} + 3\ R\text{-}COONa\ (\text{soaps})$$

Rancidity

- Rancidity is the term used to represent the deterioration of fats and oils resulting in the unpleasant taste.
- Fats containing unsaturated fatty acids are more susceptible to rancidity.
- Rancidity occurs when fats and oils are exposed to air, moisture, light, bacteria etc.
- Hydrolytic rancidity occurs due to partial hydrolysis of TAG by bacterial enzymes.
- Oxidative rancidity is due to oxidation of unsaturated fatty acids.
- Results in the formation of unpleasant products such as dicarboxylic acids, aldehydes, ketones etc.
- Rancid fats and oils are unsuitable for human consumption.

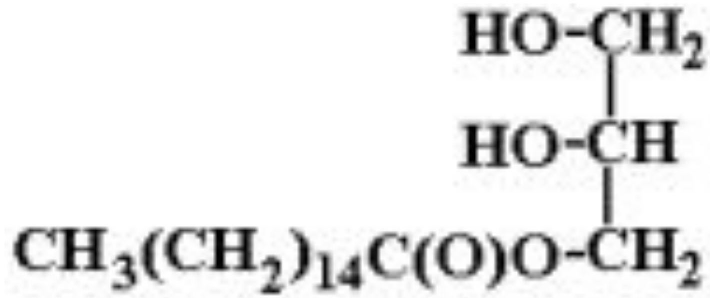
Lipid peroxidation

- Lipid peroxidation is the oxidative degradation of lipids, particularly polyunsaturated fatty acids (PUFAs) in cell membranes, caused by free radicals.
- It is fortunate that the cells possess antioxidants such as vitamin E, and superoxide dismutase (SOD) to prevent in vivo lipid peroxidation.

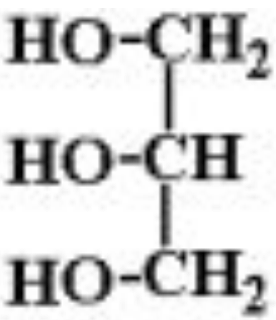


Iodine number:

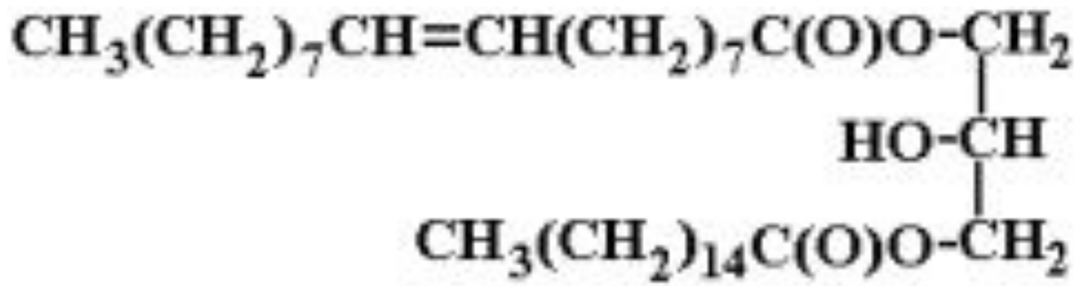
- It is defined as the grams (number) of iodine absorbed by 100 g of fat or oil.
- Iodine number is useful to know the relative unsaturation of fats, and is directly proportional to the content of unsaturated fatty acids.
- Thus lower is the iodine number, less is the degree of unsaturation.



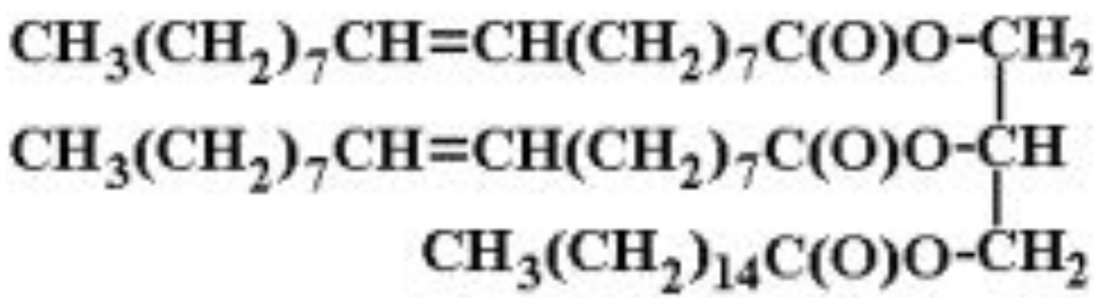
Monoglyceride



Glycerol

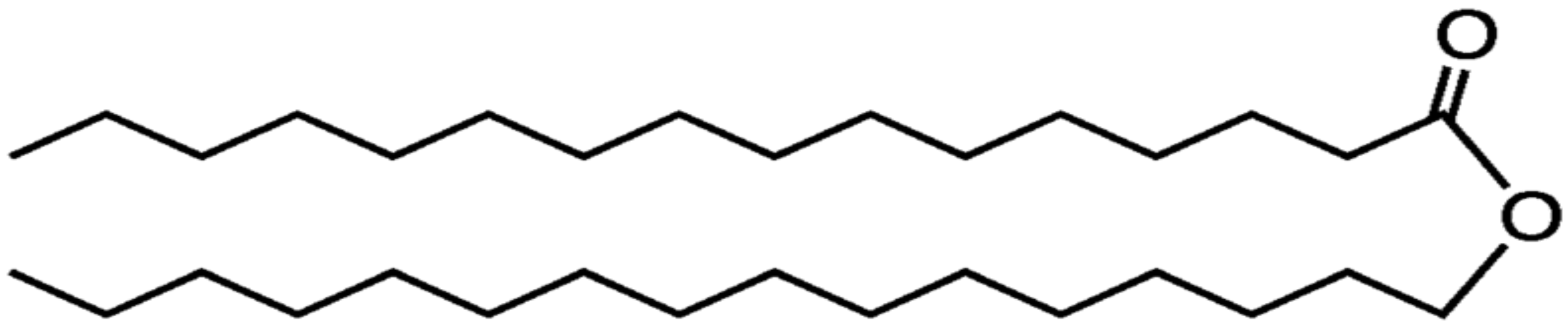


1,3 Diglyceride may be 1,2



Triglyceride

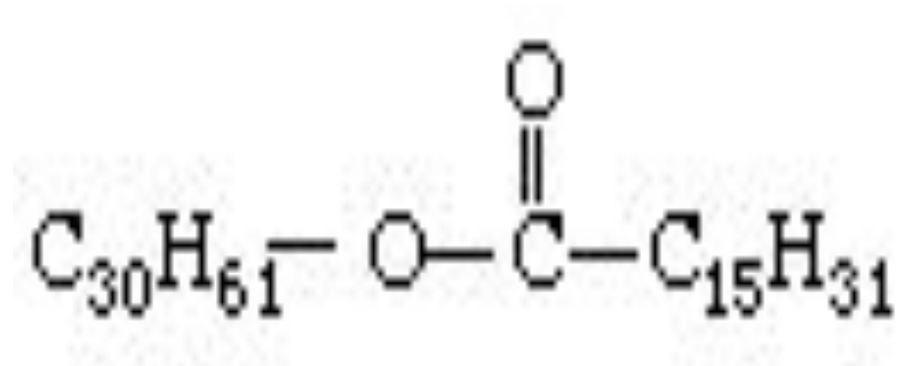
- **2:WAXES:**
- These are esters of fatty acids with alcohols other than glycerol (monohydric long chain alcohols (C16 - 30).
- Eg. Bees wax, lanolin, spermaceti oil.
- Used for lotions, ointments.



Wax (Cetyl palmitate)

- **Animal waxes:**
- **Beeswax** is a natural wax produced in the bee hive of honey bees. It is mainly esters of fatty acids and various long chain alcohols.

Beeswax (myricyl palmitate)



COMPOUND LIPIDS

- Compound lipids contain one or more additional elements, such as phosphorus, nitrogen, or sulfur along with carbon hydrogen and oxygen.
- “Esters of fatty acids with alcohol + containing additional groups (prosthetic group)”.
- Include in it

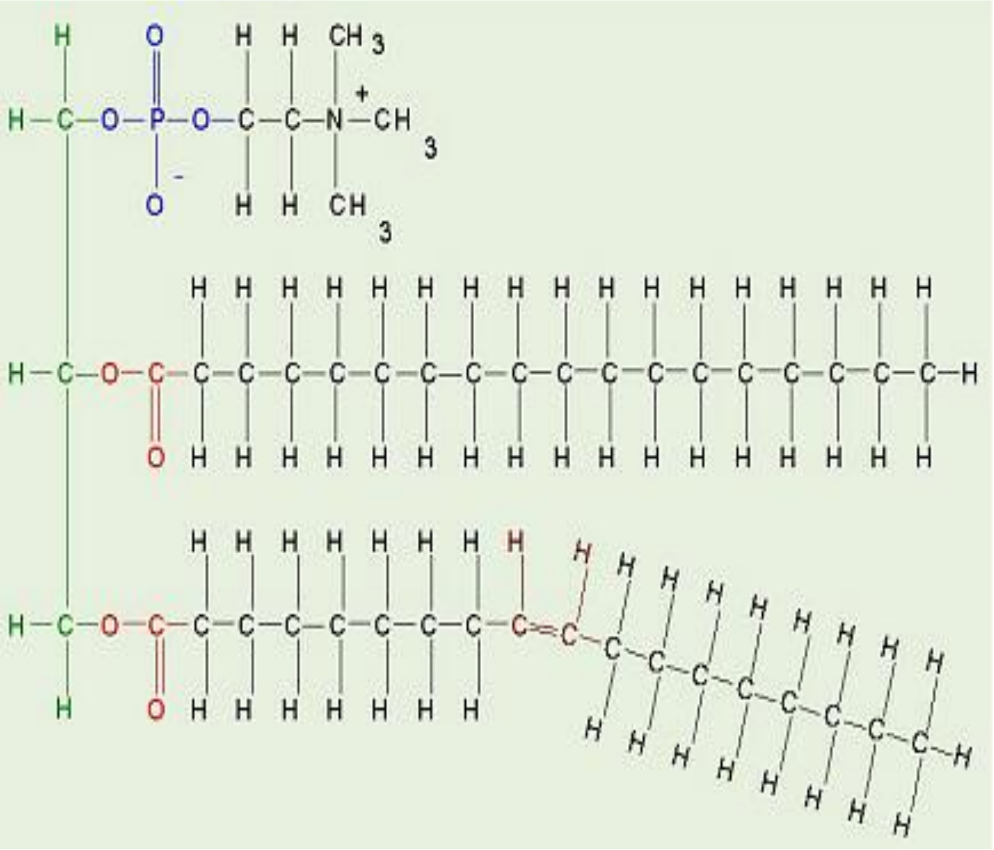
I. Glycerophospholipid

II. Sphingolipids

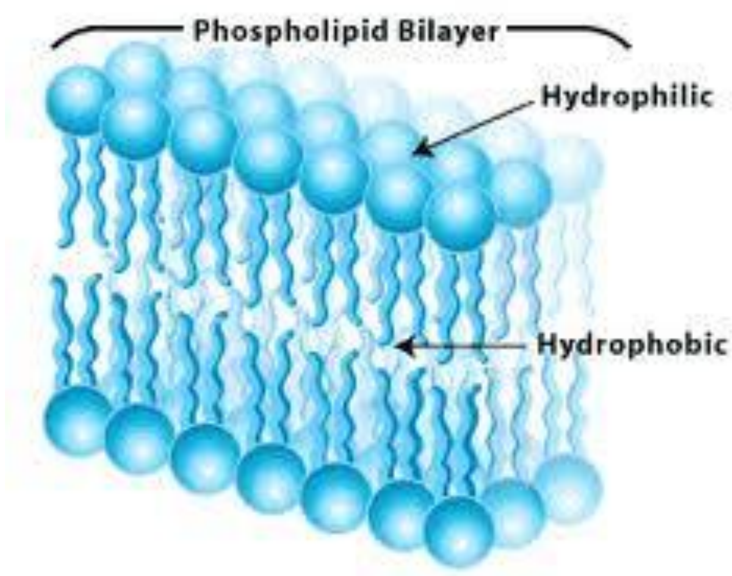
III. Lipoproteins

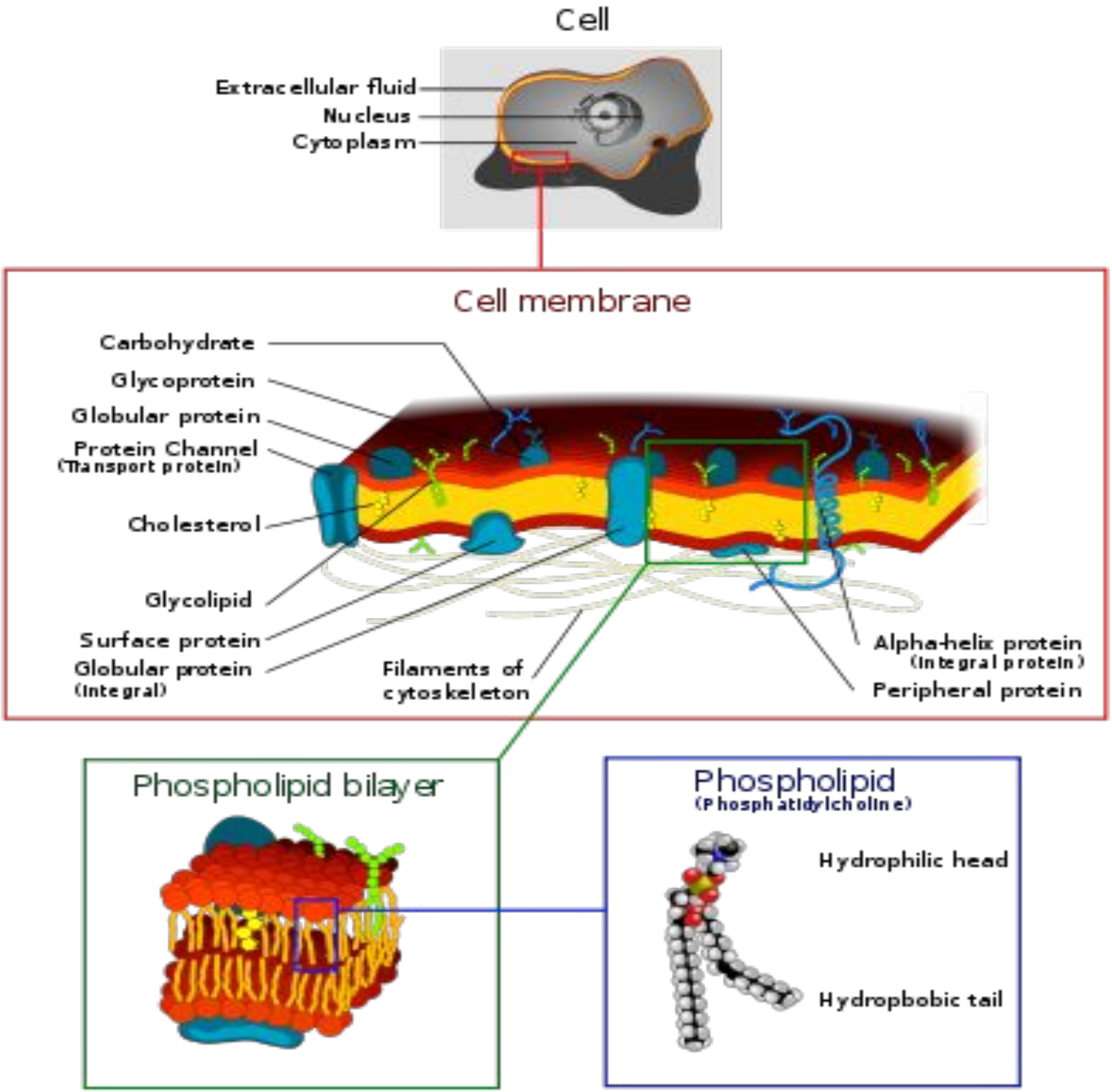
A:Glycerophospholipids

GLYCERO PHOSPHOLIPIDS are made from glycerol, two fatty acids, and (in place of the third fatty acid) a **<u>phosphate group</u>** with some other molecule attached to its other end. The hydrocarbon tails of the fatty acids are still hydrophobic, but the phosphate group end of the molecule is hydrophilic because of the oxygens with all of their pairs of unshared electrons. This means that phospholipids are soluble in both water and oil(amphiphilic or amphipathic).

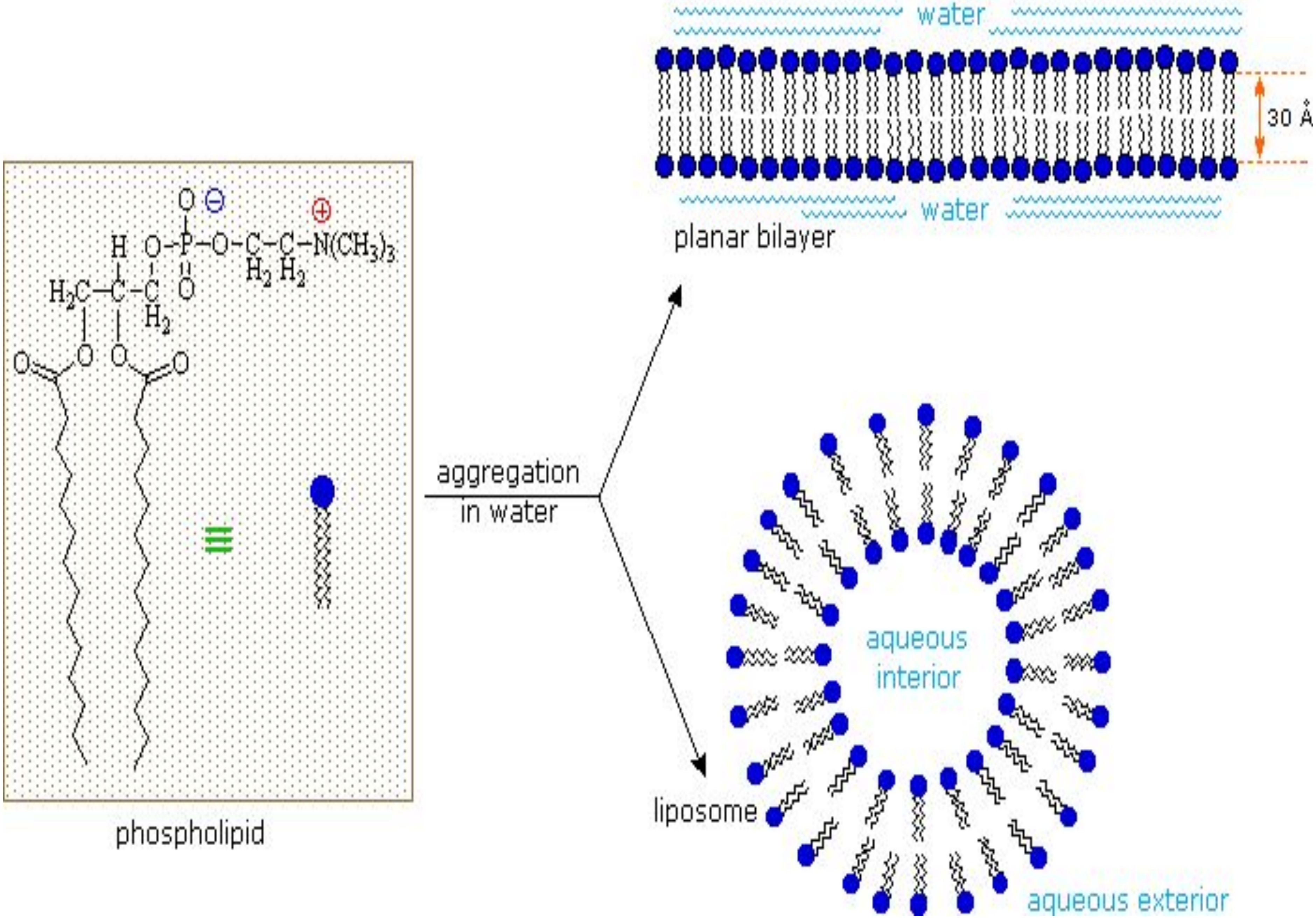


They are the main component of biological membranes. Our **cell membranes** are made mostly of phospholipids arranged in a double layer with the tails from both layers "inside" (facing toward each other) and the heads facing "out" (toward the watery environment) on both surfaces.





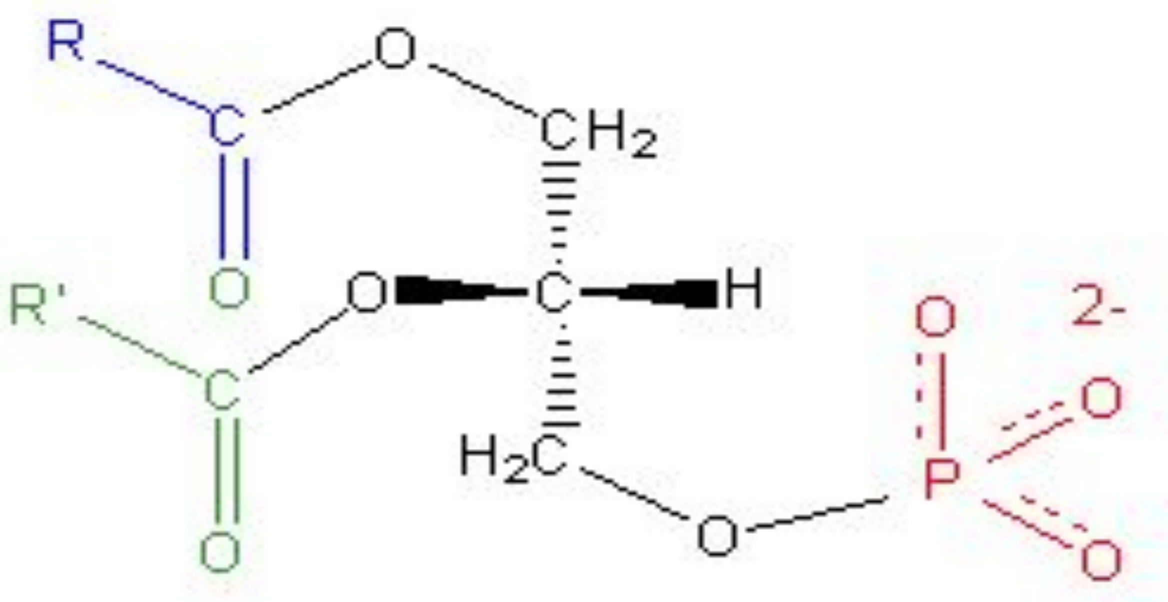
Cell
Extracellular fluid
Nucleus
Cytoplasm
Cell membrane
Carbohydrate
Glycoprotein
Globular protein
Protein Channel
(Transport protein)
Cholesterol
Glycolipid
Surface protein
Globular protein
(Integral)
Filaments of
cytoskeleton
Alpha-helix protein
(Integral protein)
Peripheral protein
Phospholipid bilayer
Phospholipid
(Phosphatidylcholine)
Hydrophilic head
Hydropbobic tail


water
30 Å
water
planar bilayer
aggregation
in water
aqueous
interior
liposome
aqueous exterior
phospholipid

Types of Phosphoglycerides or Glycerophospholipids

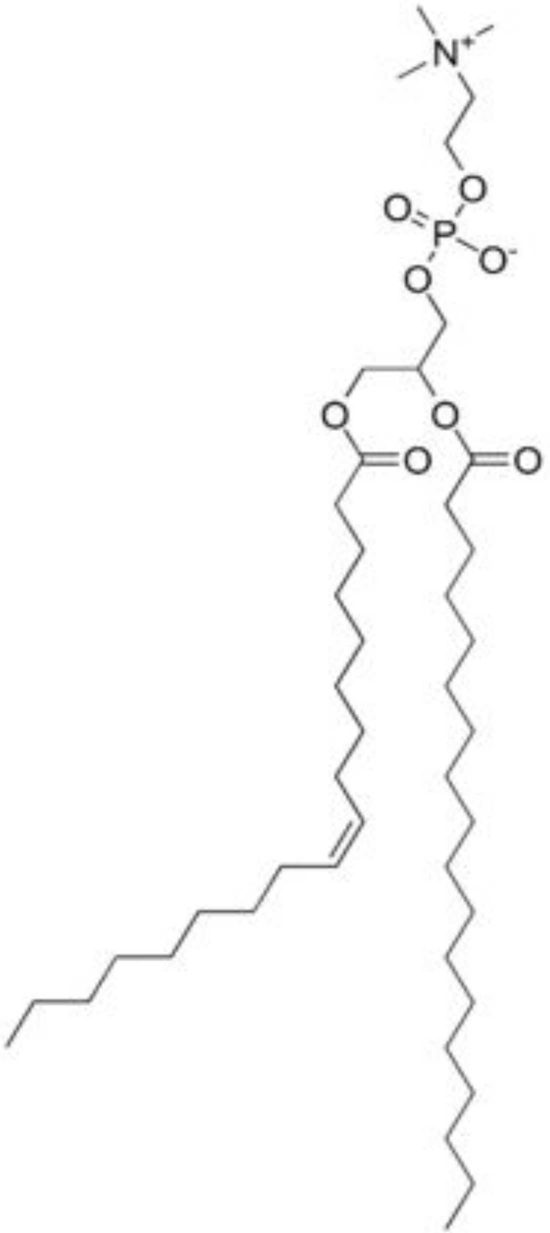
- **1:PHOSPHATIDIC ACID:**

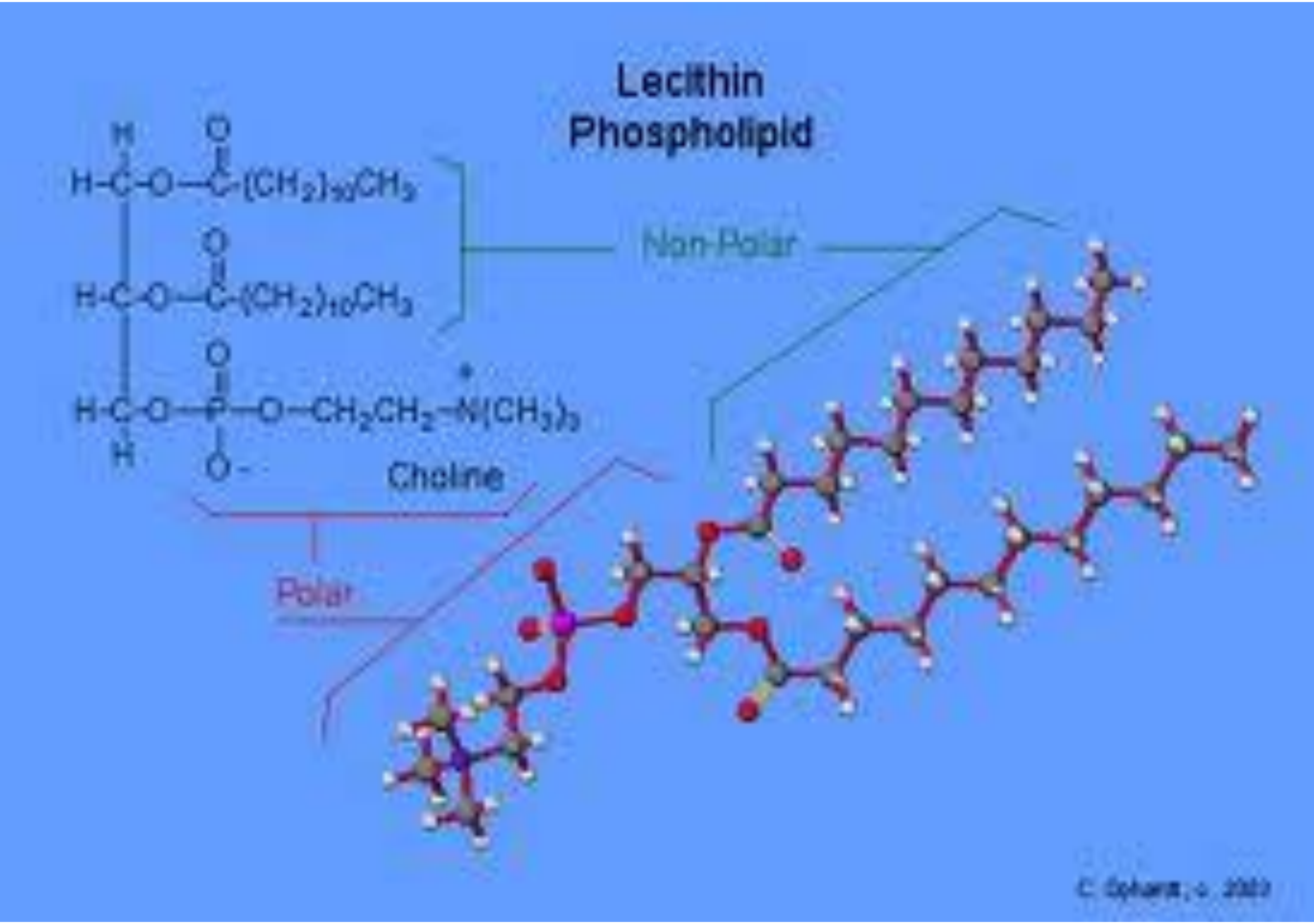
- Phosphatidic acid consists of a glycerol backbone, with, in general, a fatty acid bonded to carbon-1, an fatty acid bonded to carbon-2, and a phosphate group bonded to carbon-3.



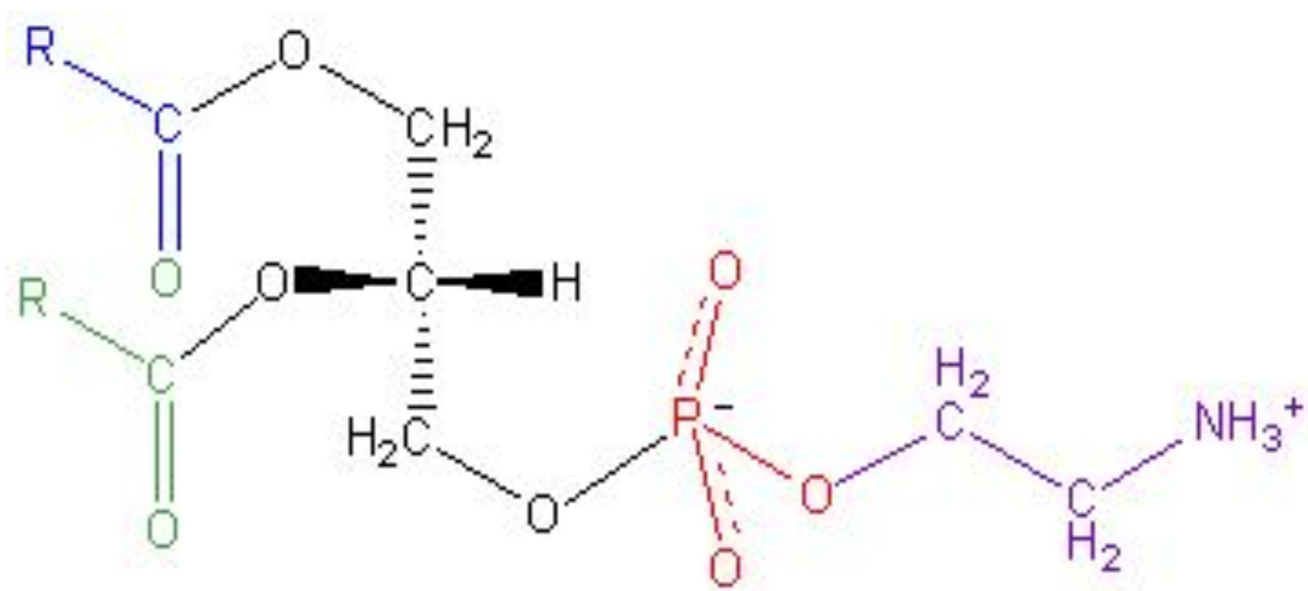
2.LECITHINS

- **Lecithin**(phosphatidylcholine) is yellow-brownish fatty substances occurring in animal and plant tissues, and in egg yolk, composed of phosphoric acid, choline(nitrogenous base), fatty acids, glycerol .




Lecithin
Phospholipid
Non-Polar
Choline
Polar

- **3:CEPHALINS:**
- **Cephalin** or **Phosphatidylethanolamine** (PE) is a phospholipid,found in biological membranes. It is synthesized by the addition of ethanolamine to diglyceride.
- Cephalin is found in all living cells, although in human physiology it is found particularly in nervous tissue such as the white matter of brain, nerves, neural tissue, and in spinal cord.

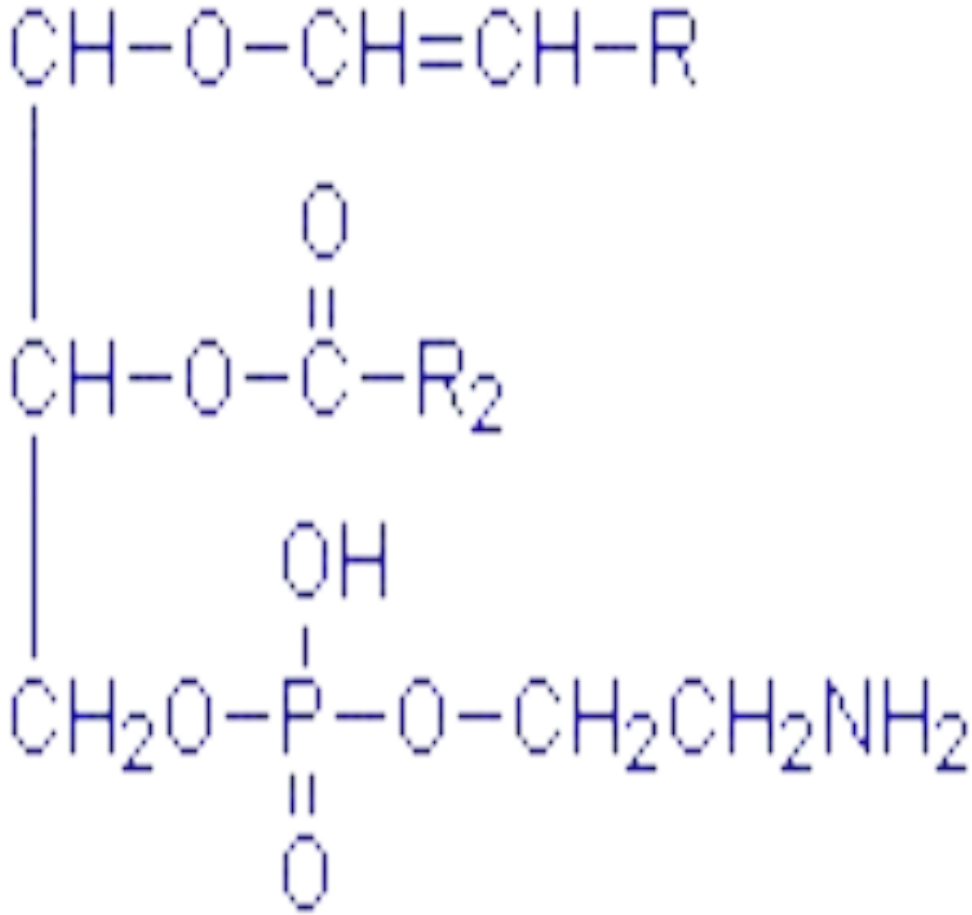


4:PLASMALOGEN

- A **plasmalogen** is an ether lipid where the first position of glycerol binds a vinyl residue (from a vinyl alcohol) with the double bond next to the ether bond.

- The second carbon has a typical ester-linked fatty acid, and the third carbon usually has a phospholipid head group like choline or ethanolamine.

- In many tissues plasmalogens are minor constituents, but in heart tissue nearly 50% of phosphatidylcholine contains the alkenyl ether at carbon 1. Nervous tissues, testes and kidneys also contain significant amounts of plasmalogens.

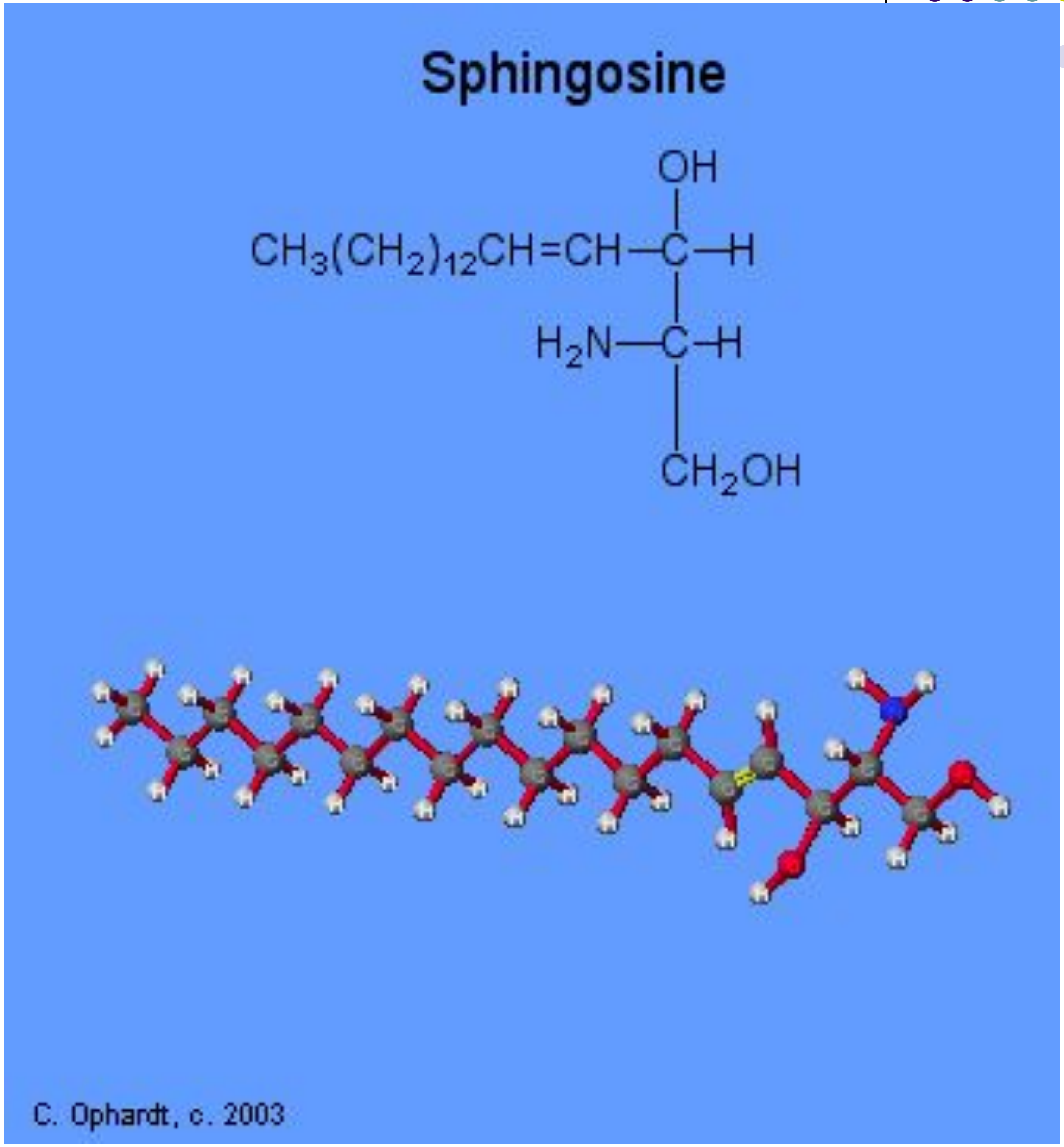


B:SPHINGOLIPIDS

Sphingolipids are a second type of lipid found in cell membranes, particularly nerve cells and brain tissues. They do not contain glycerol, but retain the two alcohols with the middle position occupied by an amine called sphingosine.

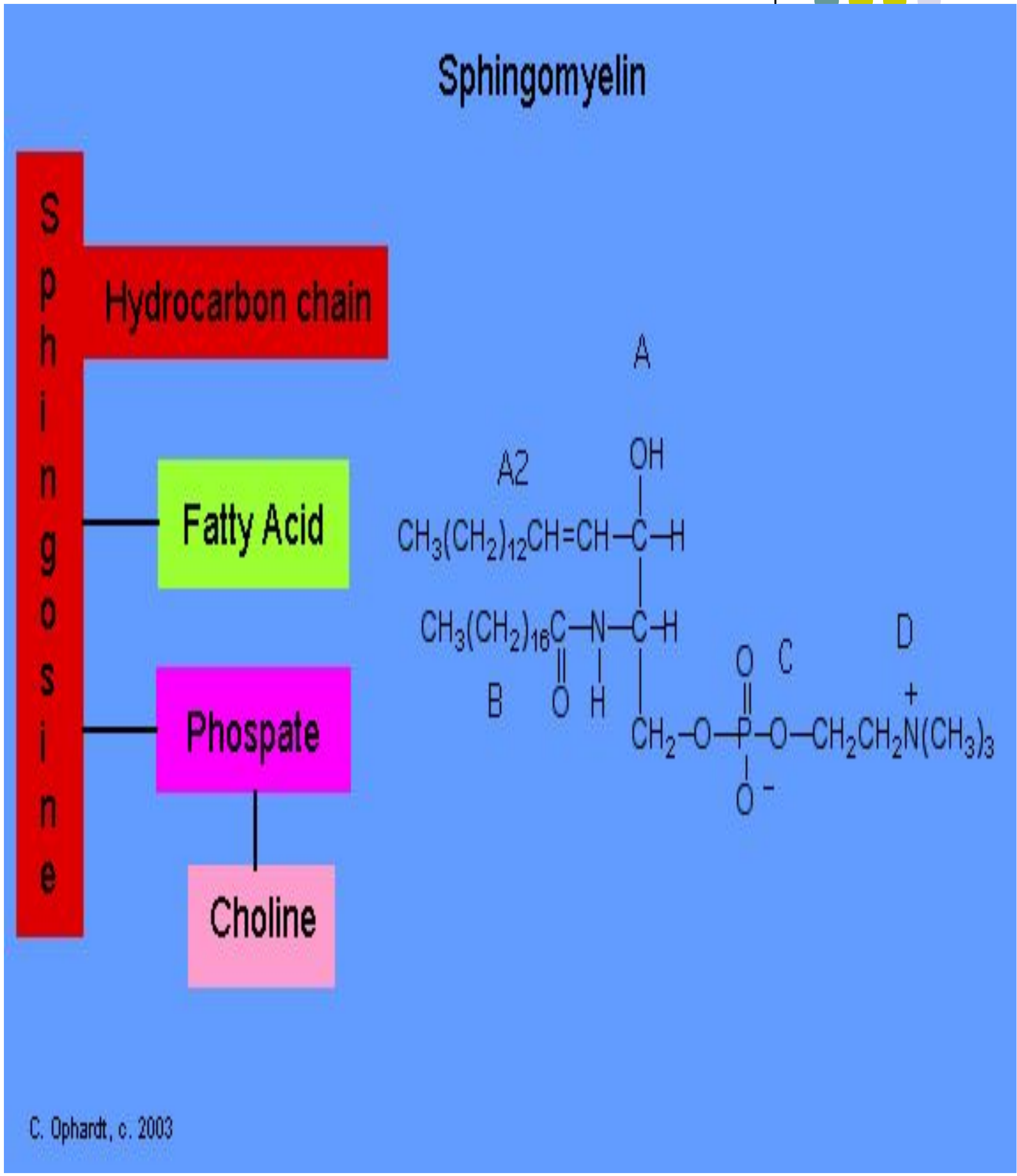
Sphingosine and fatty acid combination is called CERAMIDE.

As shown in the graphic on the left, sphingosine has three parts, a three carbon chain with two alcohols and amine attached and a long hydrocarbon chain.



TYPES OF SPHINGOLIPIDS

- **1:SPHINGOMYELINS**
- In sphingomyelin, the base sphingosine has several other groups attached.A fatty acid is attached to the amine through amide bond. Phosphate is attached through a phosphate ester bond, and again through a phosphate ester bond to choline.

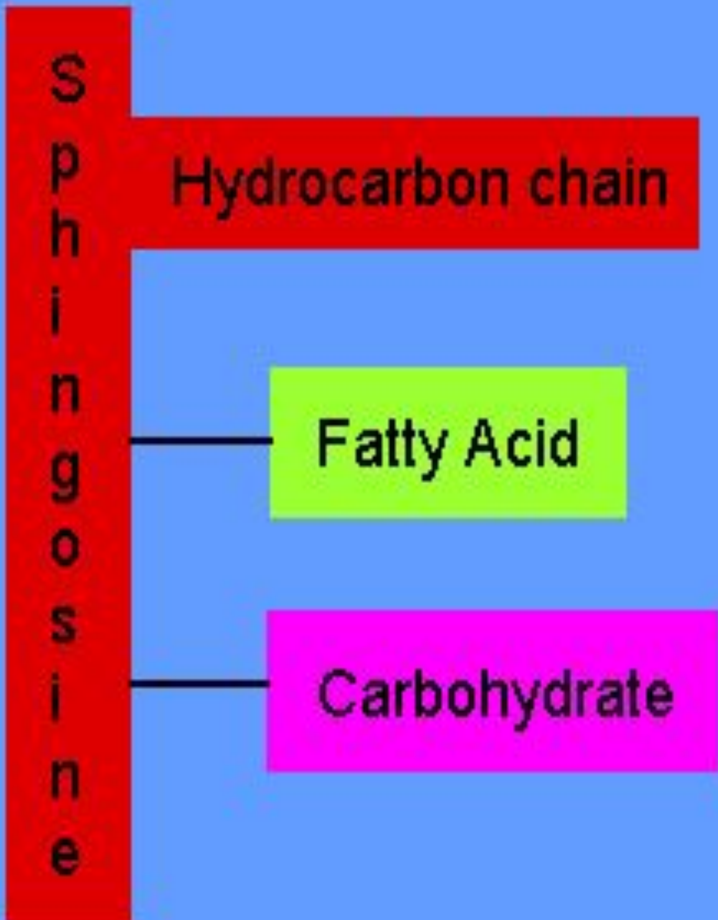


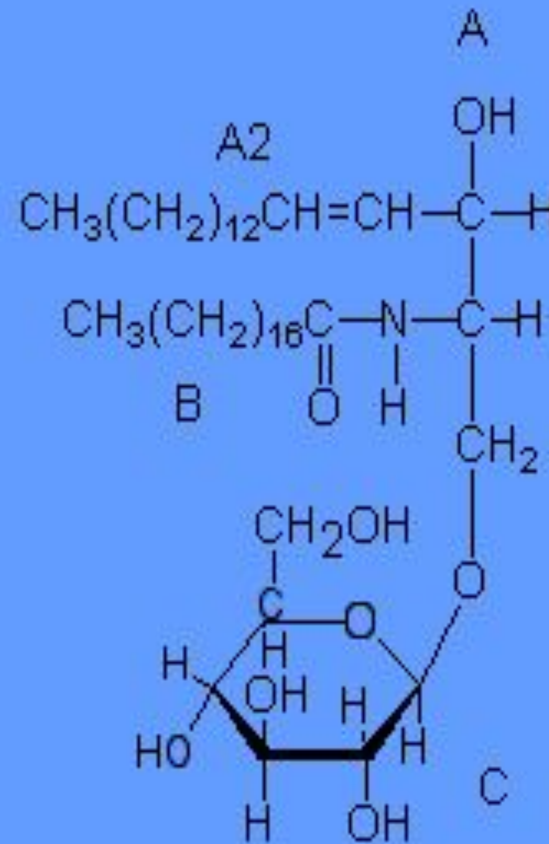
- The human brain and spinal cord is made up of gray and white regions. The white region is made of nerve axons wrapped in a white lipid coating, the myelin sheath, which provides insulation to allow rapid conduction of electrical signals.
- Sphingomyleins are located throughout the body in nerve cell membranes. They make up about 25 % of the lipids in the myelin sheath that surrounds and insulates cells of the central nervous system.

2:GLYCOLIPIDS

- Glycolipids are complex lipids that contain carbohydrates. Cerebrosides are an example which contain the sphingosine backbone attached to a fatty acid and a carbohydrate. The carbohydrates are most often glucose or galactose.
- Galactocerebroside is found almost exclusively in the membranes of brain cells.

Glucocerebroside

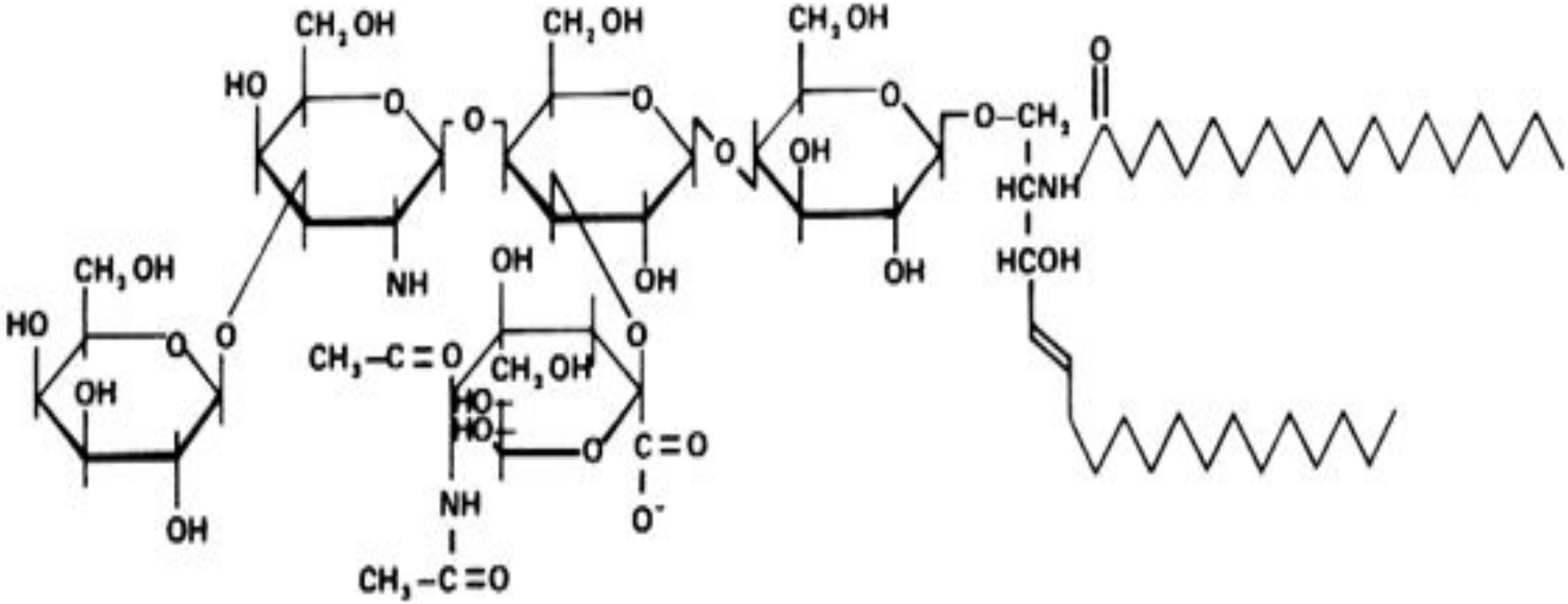


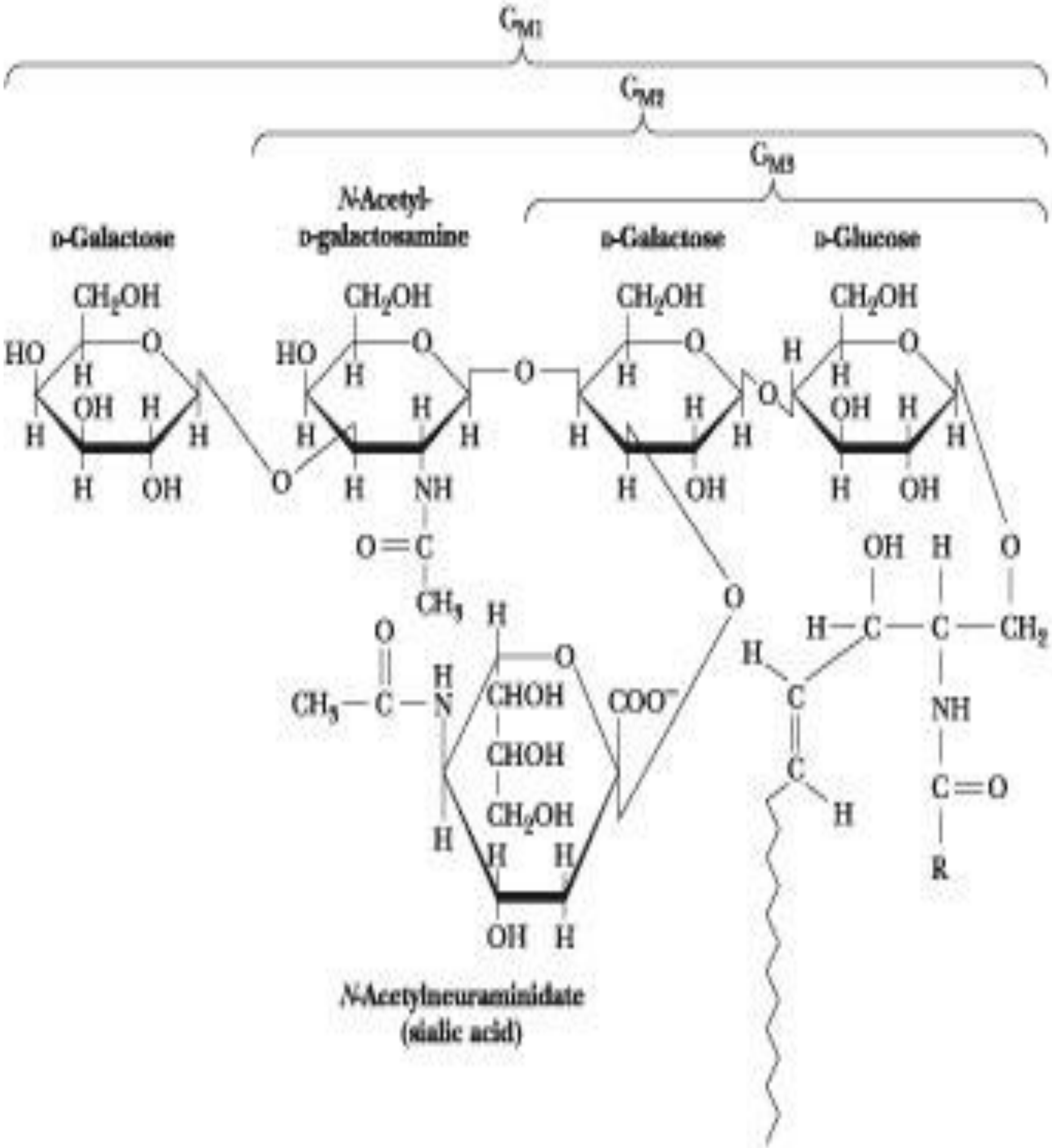


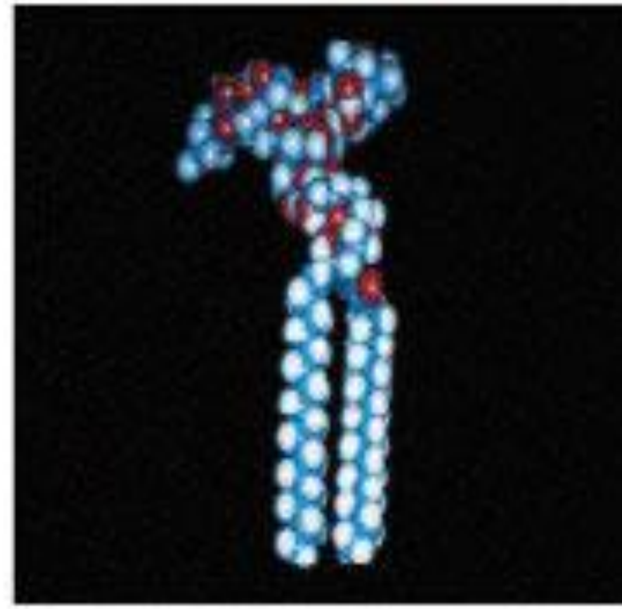
C. Ophardt, c. 2003

3:GANGGLIOSIDES:

- Gangliosides are the group of sphingolipids that show the greatest structural variation and also the more complex structure. Gangliosides are characteristic of vertebrate nervous tissues. This group includes molecules composed of ceramide linked by a glycosidic bond to an oligosaccharide chain containing hexose and **N-acetylneuraminic** acid (NANA, acidic sugar known also as **sialic acid**) units.


CH₂OH
HO
O
O
OH
NH
OH
HCNH
HCOH
CH₃—C=O
CH₂OH
HO
C=O
O⁻
CH₃—C=O
O—CH₂

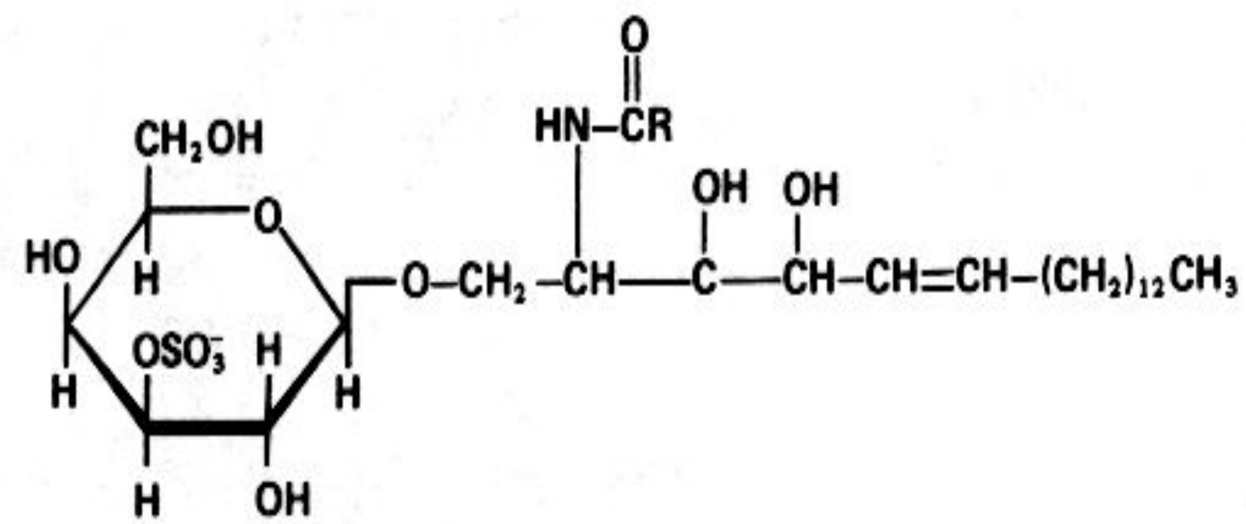




Gangliosides G_{M1}, G_{M2}, and G_{M3}

4:SULFOLIPIDS OR SULFATIDES

- **Sulfolipids** are a class of lipids which possess a sulfur-containing functional group .
- This group (known also as **cerebroside 3-sulfate**) is mainly formed of 3-sulfate esters of galactosylcerebrosides (galactosyl-3-sulfate esters) .

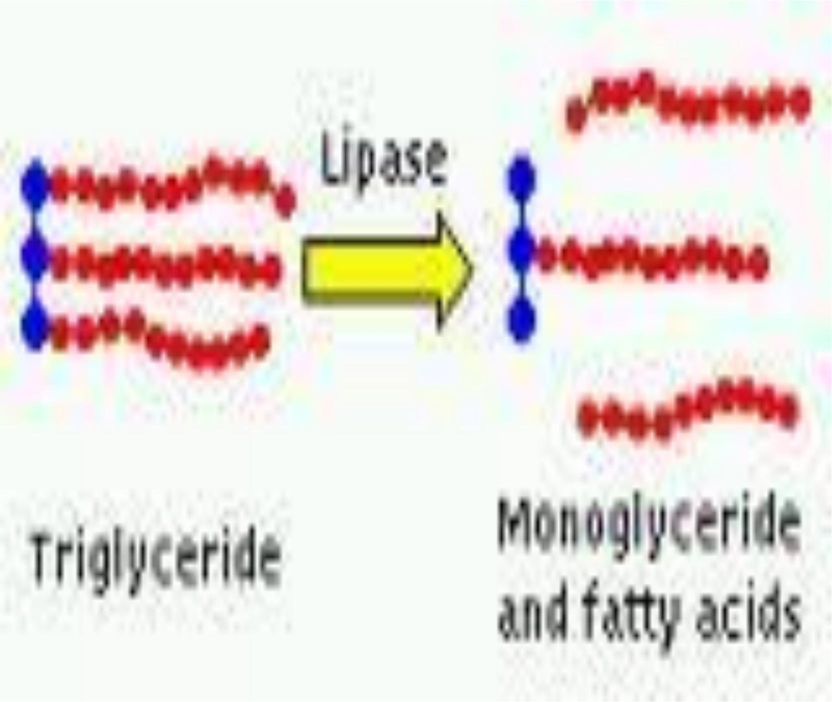


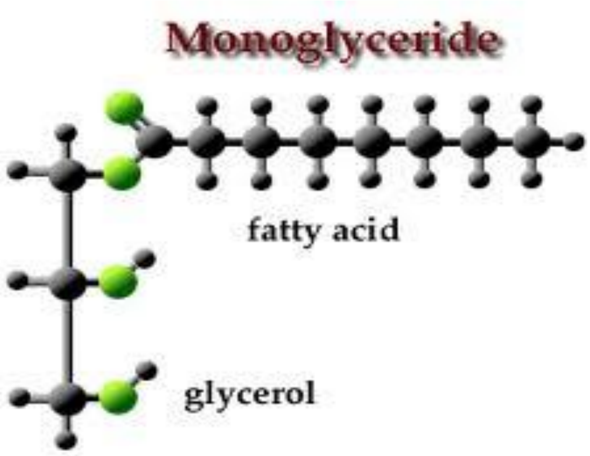
DERIVED LIPIDS

DERIVED LIPIDS

Derived from simple lipids or complex lipids on hydrolysis or these are the hydrolytic products of simple and compound lipids and include:

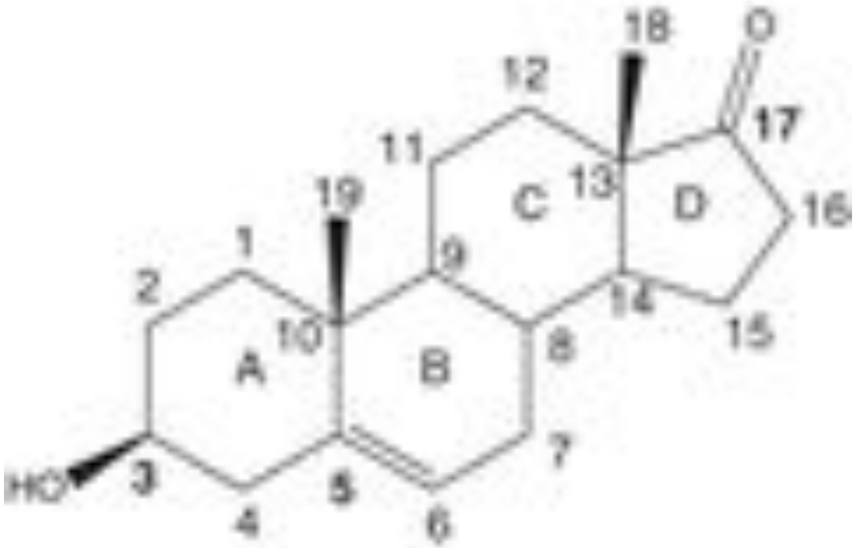
- **Fatty acids, glycerol.**
- **Mono and Diglycerides (Diacyl glycerols)**
- **Steroids- lipids with cyclopentanoperhydrophenanthrene (steroid) ring.**
  - **E.g. – cholesterol and its derivatives- bile acids, hormones, Ergosterol.**
- **Eicosanoids – Arachidonic acid derivatives- PG,TXA2**
- **Vit D, E, K**

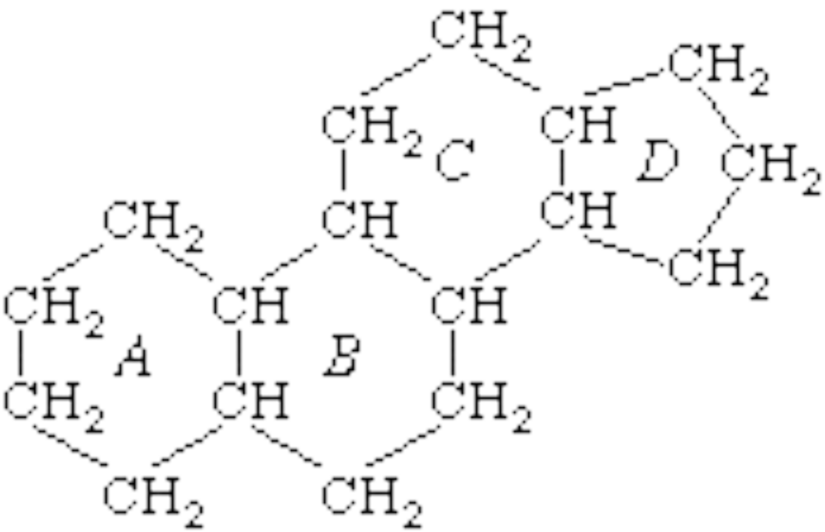

Lipase
Triglyceride
Monoglyceride
and fatty acids


Monoglyceride
fatty acid
glycerol

STEROIDS and STEROLS

- A **steroid** is a type of organic compound that contains a specific arrangement of four cycloalkane rings that are joined to each other. Examples of steroids include the dietary fat cholesterol, the sex hormones estradiol and testosterone, and the anti-inflammatory drug dexamethasone.
- Their parent nucleus is consist of 3 six-membered rings (A, B and C) and 1 five-membered ring (D) or three cyclohexane rings (designated as rings A, B, and C) and one cyclopentane ring (the D ring).
- These rings are joined or fused to each other and have a total of 17 C atom
- In most of the natural steroids a methyl group is present at C-13 and one is usually present at C-10.





gonane (1)

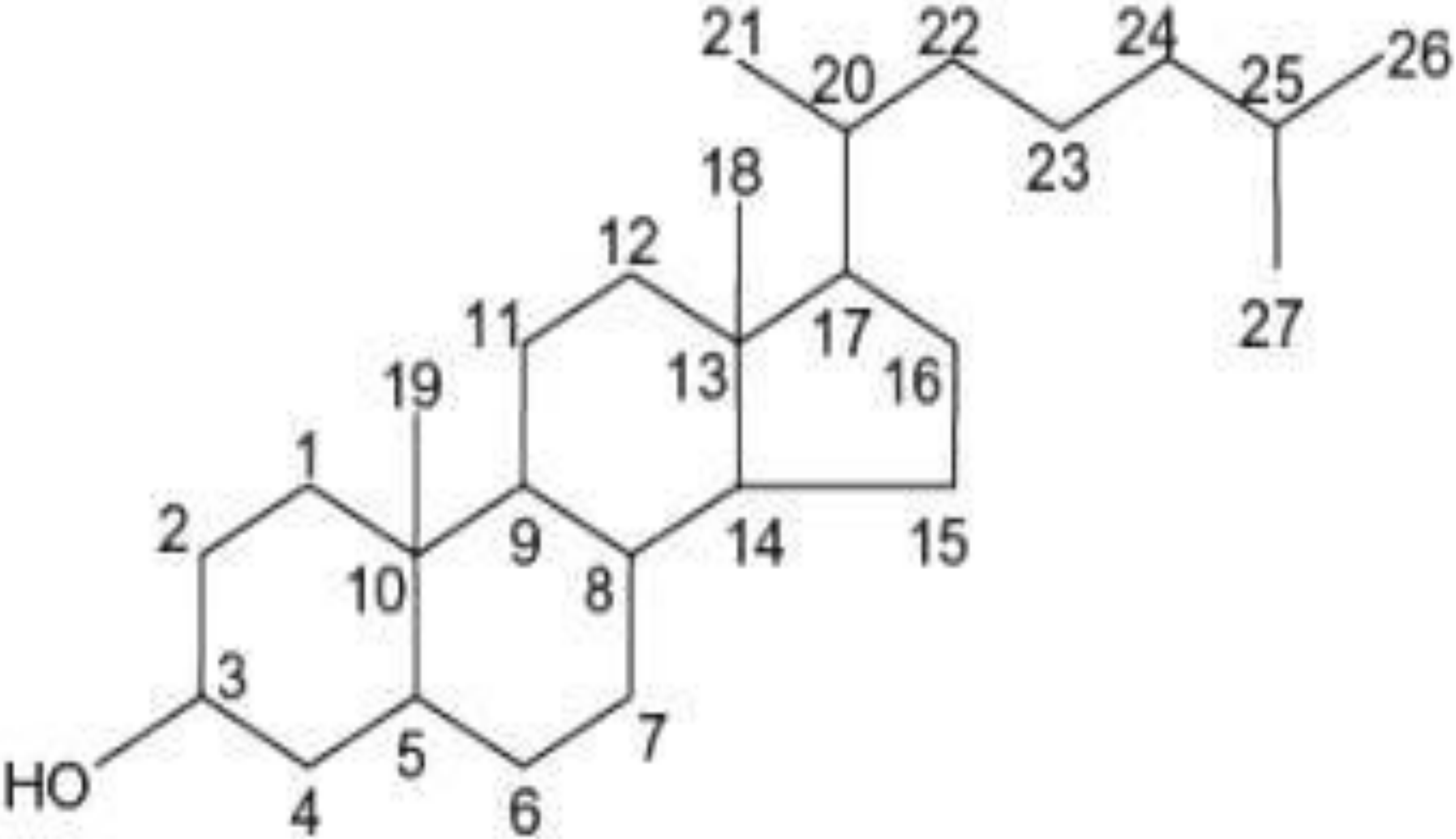
STEROLS

- Steroids with 8 to 10 carbon atoms in the side chain at C-17 and an alcohol (hydroxyl group) at C-3 are classified as sterols.

OR

- A sub group of steroids is sterols which contain a OH group at C No.3 of ring A and a branched aliphatic chain of 8 or more C atoms at C No 17 .

- Cholesterol is the major sterol in animal tissues.


HO
1
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27

CHOLESTEROL

- The name cholesterol originates from the Greek “chole- (bile) and stereos (solid), and the chemical suffix *-ol* for an alcohol.

- **Definition:**
- A compound of the sterol found in most body tissues, including the blood and the nerves. Cholesterol and its derivatives are important constituents of cell membranes and precursors of other steroid compounds, but high concentrations in the blood (mainly derived from animal fats in the diet) are thought to promote cardiovascular diseases.

OR

- A lipid unique to animal cells that is used in the construction of cell membranes and as a building block for some hormones.

FUNCTIONS OF CHOLESTEROL

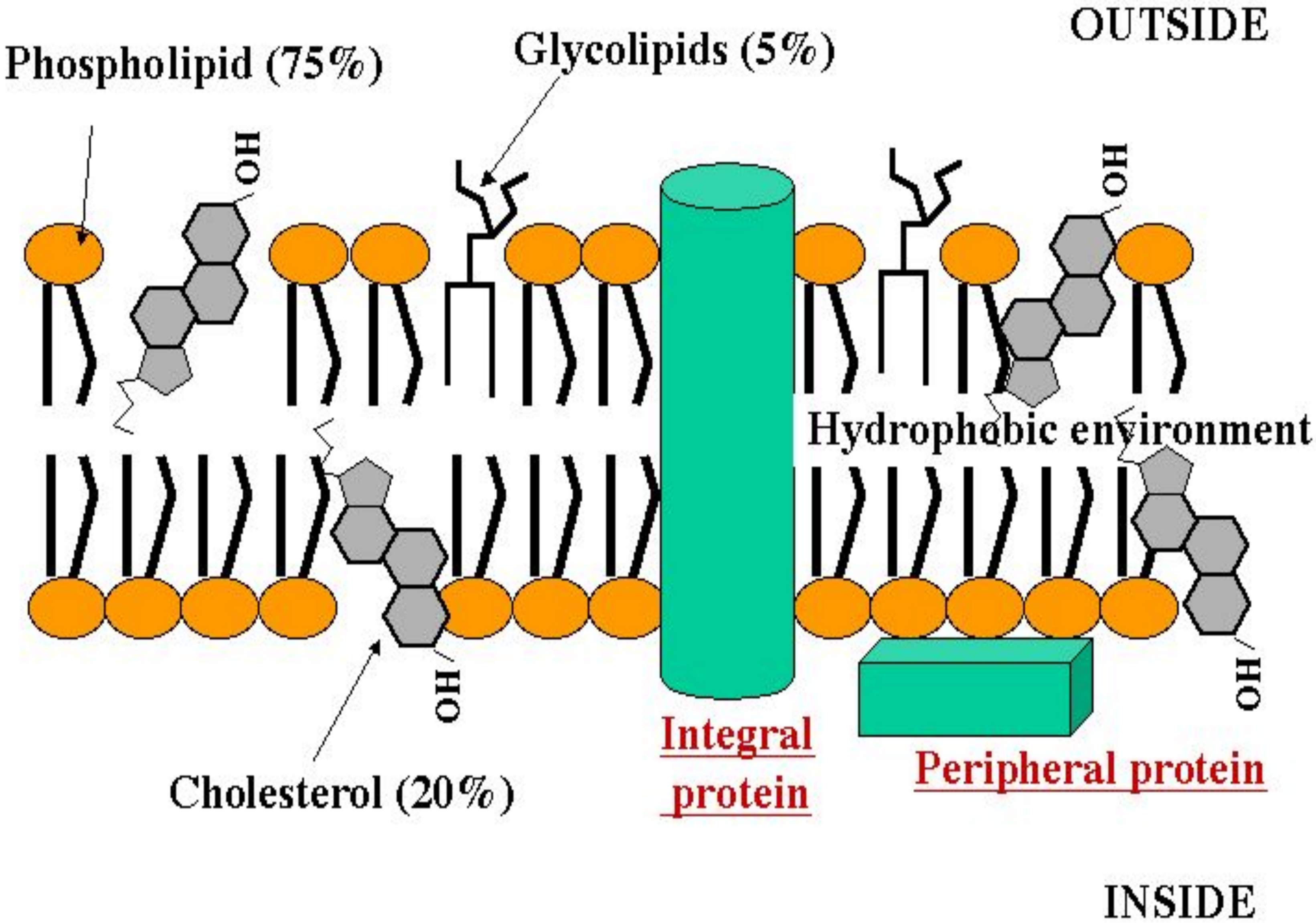
- Cholesterol is the most abundant and most common type of steroid or sterol in the human body.
- Cholesterol performs a number of essential functions in the body.

For example,

- It is essential to the formation of:
  - Bile acids (which aid in the digestion of fats)
  - Vitamin D
  - Progesterone
  - Estrogens(estradiol, estrone, estriol)
  - Androgens (androsterone, testosterone)
  - Mineralocorticoid hormones (aldosterone, corticosterone) and
  - Glucocorticoid hormones (cortisol).

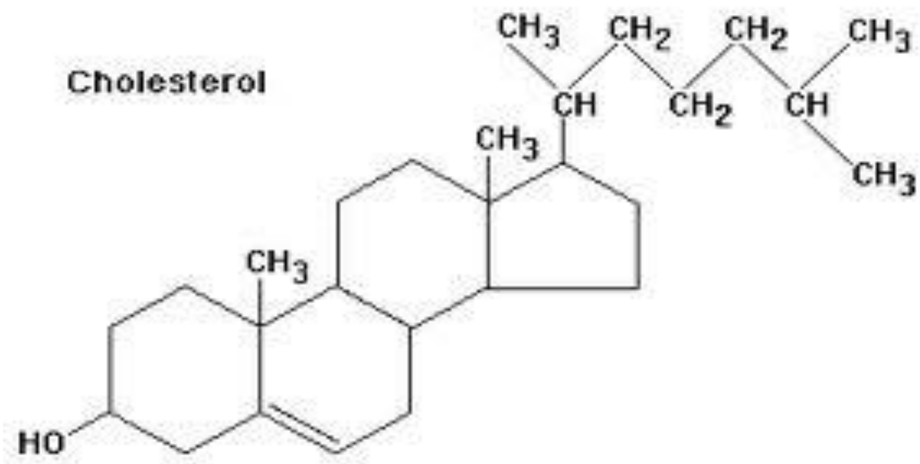
- Cholesterol is also necessary to the normal permeability and function of cell membranes.

Plasma membrane

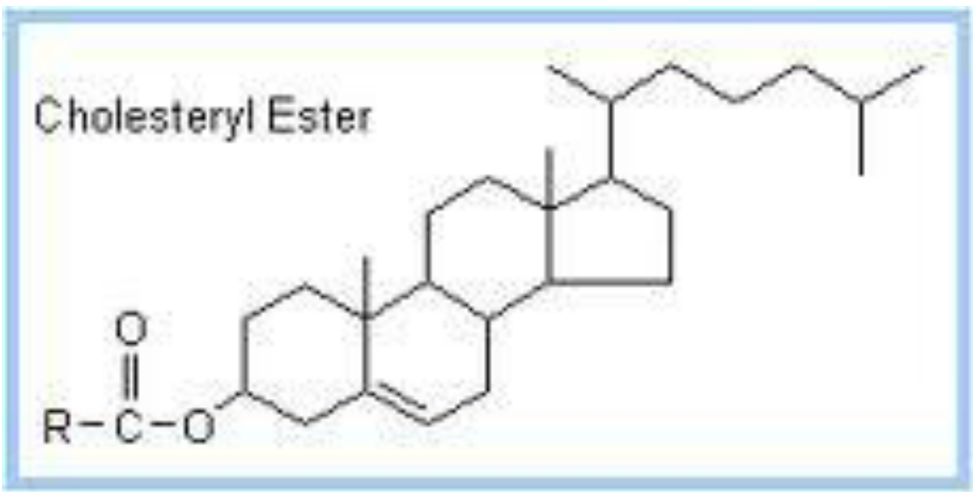


STRUCTURE OF CHOLESTEROL

- Cholesterol consist of four fused rings with the carbon membered in sequence, and an eight membered, branched hydrocarbon chain attached to D ring.
- .i:e C27H45OH



- **CHOLESTEROL ESTER:**
- If a fatty acid attached at carbon 3 of cholesterol, becomes esterified form of cholesterol which is abundantly present in plasma.



PLASMA LEVEL OF CHOLESTEROL

- Normal plasma level ranges from **150 to 240mg per 100** ml but a level of **200mg/100ml** is at present considered to be the maximum permissible.

- Typical daily dietary intake in the is 200–300 mg. The body compensates for cholesterol intake by reducing the amount synthesized.

- However, our bodies sometimes make more cholesterol than we need, and the excess cholesterol circulates in the bloodstream. High levels of cholesterol in the blood can clog blood vessels and increase the risk for heart disease and stroke

SITES OF CHOLESTEROL SYNTHESIS:

- **Although some cholesterol is obtained from the diet.**
- All tissues synthesize cholesterol but the most active ones are liver, small intestine, skin, aorta, adrenal cortex and gonads.
- Most of the cholesterol is made in the liver

Metabolism, recycling and excretion

- 80% to 90% of the cholesterol excreted from the body is lost in the form of bile acid metabolites.
- Cholesterol is oxidized by the liver into a variety of bile acids.
- These in turn are conjugated with glycine and taurine.
- A mixture of conjugated and non-conjugated bile acids along with cholesterol itself is excreted from the liver into the bile.
- Approximately 95% of the bile acids are reabsorbed from the intestines and the remainder lost in the feces.
- Under certain circumstances, when more concentrated, as in the gallbladder, cholesterol crystallises and is the major constituent of most gallstones, although lecithin and bilirubin gallstones also occur less frequently.

THANK YOU